DP Biology – C4.1

Populations and communities

- How do interactions between organisms keep population sizes in check?
- Which interactions make the populations in a community depend on one another?

SL statements C4.1.1–18. Textbook pp. 548–577. There is no HL content in C4.1.

Image: lesser flamingos, Lake Nakuru, Kenya. Syllabub, CC BY-SA 3.0.



Objectives for each class

Class	Statements	By the end of the class, you can...
1	C4.1.1–4	Define a population and estimate its size with random quadrats and with the Lincoln index
2	C4.1.5–8	Explain carrying capacity, density-dependent control and the sigmoid growth curve, and model it with duckweed
3	C4.1.9–12	Describe a community, compare competition and cooperation, and give examples of each interspecific relationship
4	C4.1.13–15	Explain how invasive species compete, describe tests for competition, and use the chi-squared test
5	C4.1.16–18	Explain predator-prey cycles, control from higher and lower trophic levels, and allelopathy and antibiotics

What is a population?

C4.1.1



Lesser flamingos, Lake Nakuru. Syllabub, CC BY-SA 3.0.

- A **population** is a group of individuals of one species that live in the same area at the same time
- Members of a population normally interbreed
- Two populations of a species are separated by **reproductive isolation**, for example by distance, mountains or water
- Individuals interact: they compete, cooperate and mate

Discussion

The flamingos at one lake form a population. Would flamingos at a lake 300 km away be part of the same population? What would you need to know to decide? Why might the two populations become more different over time? (A4.1)

Estimating population size by random sampling

- Counting every individual is usually impossible: too many, too spread out, or hidden
- Counting all of them can also disturb or damage the habitat
- So we count a **sample** and scale up to an **estimate**
- The sample must be **random**, so that every individual has an equal chance of being counted

Organisms	Method
Sessile: plants, barnacles, mussels	Random quadrats (C4.1.3)
Motile: beetles, voles, turtles	Capture-mark-release-recapture (C4.1.4)

Key idea

An estimate is never exact. A larger number of samples gives a more reliable estimate.

Why random?

If you choose where to sample, you tend to pick places with more, or fewer, of the organism. This is bias. Random numbers remove your choice, so the sample is more likely to represent the whole area.

Random quadrat sampling

C4.1.3



A 1 m quadrat on a rocky shore. M.-N. De Casamajor, Ifremer, CC BY 4.0.

1. Lay two tape measures at right angles along two edges of the area
2. Use a random number generator to give two coordinates
3. Place the quadrat at those coordinates and count the individuals inside
4. Repeat for many quadrats, then find the **mean number per quadrat**

Scaling up

Population size = mean number per quadrat × (total area / area of one quadrat). This works for sessile organisms that can be counted, both plants, such as daisies or plantains, and animals, such as barnacles and limpets. Decide in advance how to count individuals on the edge of the quadrat.

Worked example: daisies in a school field

C4.1.3

Quadrat	1	2	3	4	5	6	7	8	9	10
Daisies	3	5	0	4	2	6	1	4	3	2

- Field: 20 m × 30 m = 600 m^2; quadrat = 1 m^2
- Mean = 30 / 10 = 3.0 daisies per m^2
- Estimate = 3.0 × 600 = 1800 daisies
- Standard deviation = 1.8 (from a calculator)

Standard deviation

It shows how much the counts vary around the mean. A large SD means the daisies are patchy, and a small SD means they are spread evenly. You do not need to learn the formula: use a calculator or spreadsheet.

Activity

Invented values, for practice. A second student finds a mean of 2.4 per m^2 in the same field. Give two reasons why the estimates differ, and suggest how to make the estimate more reliable.

Capture-mark-release-recapture

1. Capture a sample, count them (**M**) and mark them harmlessly
2. Release them and allow time to mix back into the population
3. Capture a second sample (**N**) and count how many are marked (**R**)
4. Population size = (M × N) / R. This is the **Lincoln index**

Worked example

Invented values. 40 woodlice are caught and marked. A week later, 50 are caught, and 10 of these are marked. Population = (40 × 50) / 10 = 200 woodlice.

Assumption	Why it matters
Marks are not lost and do not harm the animal	Lost marks make R too small, so N is overestimated
Marked animals mix fully with the rest	Otherwise the second sample is not random
No births, deaths or migration between samples	The population must be the same at both samples

Practice: a sea turtle population

C4.1.4



Green turtle grazing on seagrass. P. Lindgren, CC BY-SA 3.0.

Year	Tagged, M	Second catch, N	Tagged in it, R
1	30	36	12
2	28	40	14
3	25	30	15

Invented values, for practice.

Activity

Turtles are tagged on a front flipper and released. 1. Use the Lincoln index to estimate the population in each year. 2. Describe the trend. 3. Turtles live for decades and swim long distances. Which assumption of the method is most likely to be broken? Explain your answer.

Carrying capacity

C4.1.5

- **Carrying capacity** is the maximum population size that an environment can support
- As a population grows, each individual gets a smaller share of resources, so there is more competition
- Births fall or deaths rise, so growth stops at the carrying capacity

Organisms	Limited resources
Animals	Food, water, territory, nesting sites, mates
Plants	Light, water, mineral nutrients, space for roots

Activity

Choose one local animal and one local plant. For each, name the resource you think most limits the population, and suggest an observation that would test your idea.

Density-dependent control by negative feedback

C4.1.6

Density-dependent factor	Why its effect increases in a dense population
Competition	Less food, water or space per individual, so fewer offspring survive
Predation	Prey are easier to find, and predators increase in number
Pathogens and pests	They pass more easily between individuals that are close together

Negative feedback

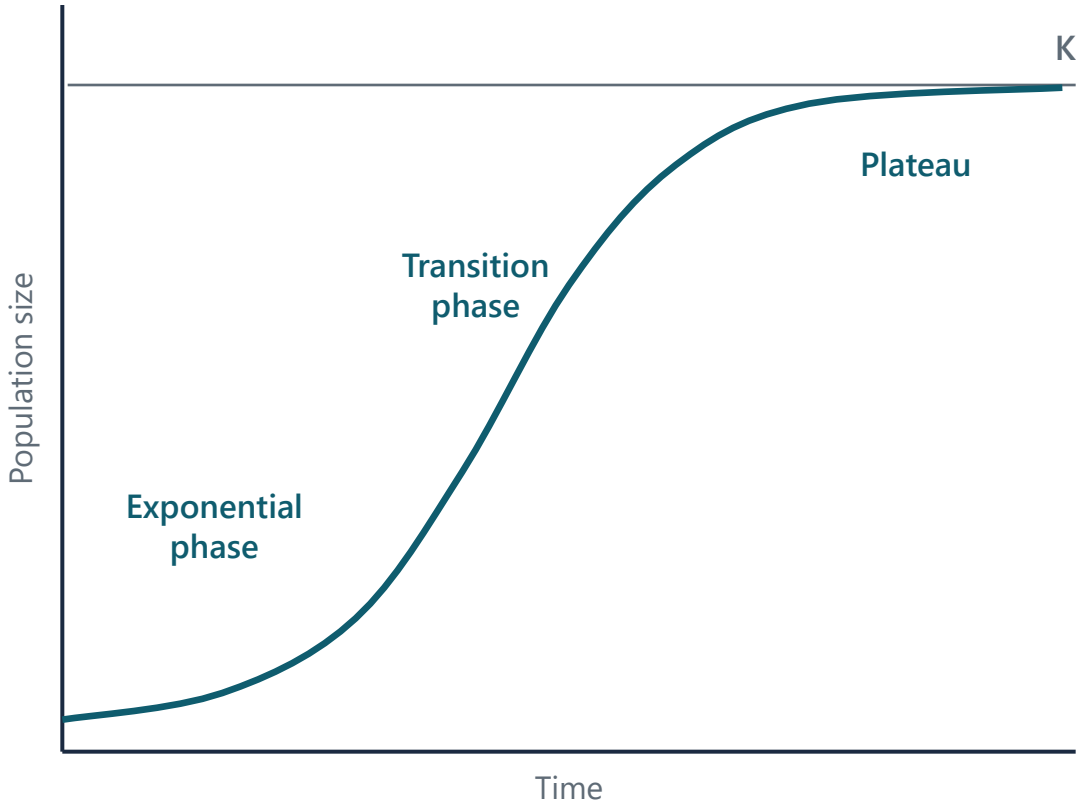
Above the carrying capacity, these factors increase deaths and reduce births, so the population falls. Below it, they relax, and the population rises again. The population fluctuates around the carrying capacity.

Density-independent factors

Floods, fires, droughts and cold winters affect a population whatever its size. They can cause sudden changes in numbers, but they do not push it back towards the carrying capacity.

The sigmoid population growth curve

C4.1.7



- **Exponential**: plenty of resources and few predators or diseases, so numbers increase by a constant proportion each generation
- **Transition**: resources run short and density-dependent factors increase, so growth slows
- **Plateau**: births and deaths balance at the carrying capacity, K

Note

A lag phase at the start is not expected here. Growth curves for real populations rarely have the smooth shape of this model.

Case study: the return of the bearded vulture

C4.1.7

- Bearded vultures were hunted to extinction in the Alps by the early 1900s
- Captive-bred birds have been released in the Alps since 1986, and in Switzerland since 1991. The first chick hatched in the wild in Switzerland in 2007
- With protection and plentiful food, the population is now growing

Why exponential at first?

A few individuals in a large area with plenty of resources face little competition. Each breeding pair adds young, and those young breed in turn, so growth speeds up. Some density-dependent factors, such as predators or competitors, may be missing.

Activity

Look up the breeding bird index of a Swiss species at vogelwarte.ch. Sketch its curve. Which phase of the sigmoid curve is it in? Suggest which resource will set its carrying capacity.

Modelling sigmoid growth with duckweed

C4.1.8



Lemna minor. Karelj, CC BY-SA 3.0.

1. Put a few duckweed plants in a beaker of pond water or nutrient solution in light
2. Count the fronds every day or two until the surface is covered
3. Plot number of fronds against time

Testing the model

Plot the number of fronds on a log scale and time on a normal scale. Exponential growth gives a straight line. Where the line curves away and levels off, growth is no longer exponential.

Extension

Change one variable, such as nutrient concentration or the surface area of the container. Predict how it will change the carrying capacity, then test your prediction.

A community: all the interacting organisms

C4.1.9

- A **community** is all the populations of different species living and interacting in an area
- It includes plants, animals, fungi and bacteria
- Together with the abiotic environment, it forms an **ecosystem** (C4.2)



A coral reef community. Jim Maragos, USFWS, public domain.

Activity

List ten organisms in a community near you, such as a forest, a pond or a meadow. Include at least one fungus and one bacterium. Draw lines between organisms that interact, and label each line with the type of interaction.

Competition and cooperation within a species

C4.1.10



Spotted hyenas. S. Jurvetson, CC BY 2.0.



Leafcutter ants. Kathy and Sam, CC BY 2.0.

Interaction	Examples
Competition	Members of a species share the same needs, so they compete for food, light, territory or mates: wildebeest males fight for territory; tree seedlings compete for light

Discussion

Competition harms both individuals, as each gets less. Cooperation can benefit all. Why then is competition never removed completely, even in a cooperative group? In social insects, why might workers that never breed still pass on their genes?

Relationships between species

C4.1.11

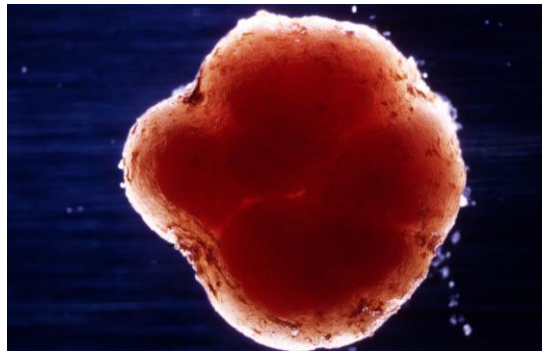
Relationship	What happens	Example
Herbivory	A primary consumer feeds on a producer	Bison grazing on grass
Predation	A predator kills and eats its prey	Lynx eating snowshoe hares
Interspecific competition	Two species use the same limited resource	Lions and leopards hunting the same prey
Mutualism	Both species benefit	Rhizobium in clover root nodules
Parasitism	A parasite lives on or in a host and harms it	Ticks feeding on deer
Pathogenicity	A pathogen lives in a host and causes disease	Chytrid fungus in the evergreen robber frog

Activity

Find a local example of each relationship. Which relationship is hardest to tell apart from another, and why?

Mutualism: both species benefit

C4.1.12







Partners	Benefit to the plant or coral	Benefit to the partner
Legume and Rhizobium	Rhizobium fixes nitrogen into ammonium, used to make amino acids	Sugars from photosynthesis, and a low-oxygen home in the nodule
Orchid and mycorrhizal fungus	Seeds have no food store, so the fungus supplies sugars, minerals and water	Adult orchids later supply carbon compounds
Hard coral and zooxanthellae	Algae supply sugars, amino acids and oxygen from photosynthesis	Shelter in shallow sunlit water, plus carbon dioxide and nutrients

Images: a cut root nodule, red from leghaemoglobin (CSIRO, CC BY 3.0); lady's slipper orchid (Johan N, public domain); coral reef (J. Maragos, public domain).

Endemic and invasive species

C4.1.13



Red squirrel. Jojo, CC BY-SA 3.0.

- **Endemic** species occur naturally in an area. **Invasive** species were introduced by humans and spread
- Many invasive species thrive because the predators, pathogens and competitors of their native range are missing
- If an invasive species is better at taking a shared resource, the endemic species declines



Grey squirrel. PierreSelim, CC BY 3.0.

Example

Grey squirrels were brought to Britain from North America in the late 1800s. They eat seeds more efficiently than red squirrels, including acorns, which red squirrels digest poorly, and they can store more fat for winter. They also carry squirrel pox, which kills reds. Red squirrels have disappeared from most of England and Wales.

Invasive species in Switzerland

C4.1.13



Nutria. G. Chernilevsky, public domain.



Himalayan balsam. R. Flogaus-Faust, CC BY 4.0.

Species	Competitive advantage
Nutria (coypu)	Escaped from fur farms; burrows into river banks and eats large amounts of waterside vegetation, taking food and space from native species
Himalayan balsam	Grows up to 2 m in a single season and shades out native plants along rivers; its exploding seed pods spread seeds several metres
Japanese knotweed	Spreads by rhizomes and outgrows native plants on river banks

Activity

Choose one invasive species near you. Explain how it outcompetes an endemic species for one named resource.

Testing for interspecific competition

C4.1.14

Approach	Method	Example
Laboratory experiment	Grow two species alone and together under the same conditions	Two Paramecium species: one excluded the other when grown together
Field observation	Random sampling to see whether two species occur together less often than expected	Chi-squared test on quadrat data (C4.1.15)

Common mistake

If one species does better when the other is absent, competition is indicated but not proven. The two species might simply need different conditions. Connell found that Chthamalus survived much better on the middle and lower shore when Semibalanus was removed, but on the upper shore removing Semibalanus made little difference.

Chi-squared: are two species associated?



Ling heather. jensu, CC0.



Bracken. L. A. Ratledge, CC BY 4.0.

- Record only the **presence or absence** of each species in many random quadrats
- H0: the two species are distributed independently
- H1: the two species are associated, positively or negatively

Observed (100 quadrats)	Bracken present	Bracken absent	Total
Ling present	18	42	60
Ling absent	32	8	40
Total	50	50	100

Invented values, for practice.

Step 1

Put the observed counts and the totals in a contingency table.

Calculating chi-squared

Expected	Bracken present	Bracken absent
Ling present	30	30
Ling absent	20	20

- **Step 2**: expected = (row total × column total) / grand total. For example, 60 × 50 / 100 = 30
- **Step 3**: for each cell, find $(O - E)^2 / E$, then add them: 4.8 + 4.8 + 7.2 + 7.2 = 24.0
- **Step 4**: degrees of freedom = (rows − 1) × (columns − 1) = 1, so the critical value at p = 0.05 is 3.84

Step 5: conclusion

24.0 is greater than 3.84, so reject H0. The two species are associated. Ling and bracken occur together less often than expected, which is a negative association. This could be caused by competition, or by allelopathy from bracken, but it could also mean that the species need different soils. The test shows an association, not its cause. Remember: degrees of freedom are 1 only for a table with two rows and two columns.

Practice: two seaweeds on a rocky shore

C4.1.15

Observed (80 quadrats)	P present	P absent	Total
Q present	24	8	32
Q absent	16	32	48
Total	40	40	80

Invented values, for practice.

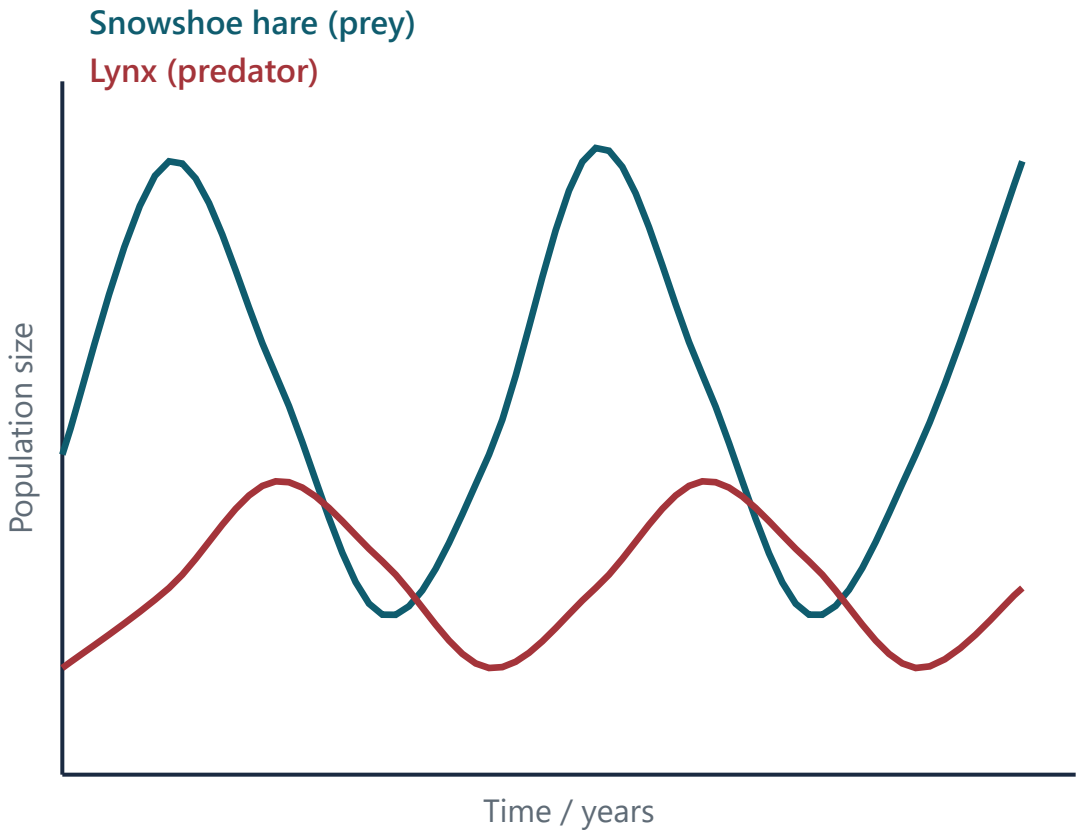
1. State H0 and H1
2. Calculate the four expected values
3. Calculate chi-squared
4. Compare it with 3.84
5. State your conclusion

Activity

Work in pairs. Then suggest one ecological reason for the result you find. Plan a removal experiment that could test whether the association is caused by an interaction between the two species, and name one variable you would need to control.

Predator-prey cycles

C4.1.16



General shape only. Peaks repeat roughly every 10 years.

1. More hares means more food for lynx, so lynx numbers rise
2. More lynx eat more hares, so hare numbers fall
3. With less food, lynx numbers fall, and hares recover



Lynx. USFWS, public domain.



Snowshoe hare. A. Schmierer, CC0.

Top-down control and bottom-up control

C4.1.17

Control	How it works	Example
Top-down	A higher trophic level limits the one below through predation or herbivory	More predatory fish → fewer forage fish → more zooplankton → less phytoplankton
Bottom-up	The supply of food at a lower level limits the levels above	More nutrients → more phytoplankton → more zooplankton → more fish



Green turtle. P. Lindgren, CC BY-SA 3.0.

Key idea

Both kinds of control can act in the same community, but one is usually dominant. Green turtles graze seagrass: the amount of seagrass limits the turtles (bottom-up), and where turtles are protected, their grazing can limit the seagrass (top-down). Removing a top predator, such as wolves, shows top-down control clearly.

Allelopathy and antibiotics

C4.1.18





Penicillium colonies. Atalyssa, CC BY 4.0.

	Allelopathy	Antibiotics
What	A plant releases chemicals that stop other plants germinating or growing	A fungus or bacterium releases chemicals that kill or slow the growth of bacteria
Example	Bracken, walnut trees and sunflowers. A local example: walnut	Penicillium makes penicillin, discovered by Alexander Fleming in 1928
Effect	Fewer competitors for light, water and minerals	Fewer competitors for food

Similarity

In both, an organism releases a chemical into its environment that deters competitors. Image above left: bracken. L. A. Ratledge, CC BY 4.0.

Check-in: populations and communities

C4.1.1–20

1. Define a population.
2. Explain why random sampling is used to estimate population size.
3. In a capture-recapture study, 40 are marked, 50 are caught next time and 8 are marked. Estimate the population.
4. Explain how density-dependent factors limit population growth.
5. Describe the three phases of a sigmoid growth curve.
6. Distinguish between mutualism, parasitism and competition.

How to use this

Answer without your notes, then compare with a partner.

Working with data: quadrat sampling

C4.1.6

Dandelions counted in ten random 1 m^2 quadrats in a 600 m^2 field. These data are invented for practice.

Quadrat	1	2	3	4	5	6	7	8	9	10
Dandelions	3	0	5	2	4	1	0	6	3	2

1. Calculate the mean number per quadrat.
2. Estimate the population in the field.
3. Explain why more quadrats would give a better estimate.
4. Suggest why the estimate could be wrong if dandelions grow in clumps.

Common mistakes to avoid

Not this	But this
Carrying capacity is the largest population ever reached.	It is the maximum population the environment can support long term.
In capture-recapture, divide by the total caught.	Divide by the number of marked individuals recaptured.
A significant chi-squared result proves causation.	It shows an association, not a cause.
Invasive species are any non-native species.	Invasive species spread and harm native communities.

Check that you can...

1. **Estimate** population size using quadrats and capture-mark-release-recapture
2. **Explain** carrying capacity and density-dependent control
3. **Describe** and **explain** the sigmoid growth curve
4. **Outline** intraspecific and interspecific relationships, including mutualism and invasive species
5. **Apply** the chi-squared test to species association
6. **Explain** predator-prey cycles, control from above and below in food webs, and allelopathy

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. 20 snails are marked and released. Later, 30 are caught, and 6 of them are marked. Estimate the population.

2. Explain why predation is a density-dependent factor.

3. State one benefit to each partner in the mutualism between legumes and Rhizobium.

Image credits

Large number of flamingos at Lake Nakuru.jpg – Syllabub – CC BY-SA 3.0

Quadrats sur le site Abbadia (Ifremer 00380-49168 – 1942).jpg – Marie-Noelle De Casamajor – CC BY 4.0

Green Sea Turtle grazing seagrass.jpg – P.Lindgren – CC BY-SA 3.0

Lemna minor Prague 2012 1.jpg – Karelj – CC BY-SA 3.0

Coral reef fish pacific blue tan paracanthurus hepatus.jpg – Jim Maragos, U.S. Fish and Wildlife Service – Public domain

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Calluna vulgaris 108500153.jpg – jensu – CC0

Common bracken (Pteridium aquilinum).jpg – Lee Ann Ratledge – CC BY 4.0

Canada lynx (6002291242).jpg – USFWS Mountain-Prairie – Public domain

SNOWSHOE HARE (Lepus americanus) (5-28-2015) quoddy head, washington co, maine -02 (18988733919).jpg – ALAN SCHMIERER – CC0

P.polonicum.jpg – Atalyssa – CC BY 4.0

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